

Green Building Design: Final Project

ASCE3105/ ENVE4105

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Group 15

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Integrated Design Process

The objective of this project is dedicated to creating a south-facing façade tailored for energy-conscious Ottawa residents who prioritize sustainable living solutions. The goal of our project is to design a home façade and assembly that seamlessly integrates solar energy utilization. This system is designed to harness solar energy effectively and offset approximately 45% of net energy used per year.

Ottawa's climate presents a unique challenge, with harsh winters characterized by average January lows of approximately -14°C [12]. These conditions make climate-adaptive housing solutions a necessity rather than a luxury. Residents face increasing energy costs due to prolonged heating demands during winter, making energy-efficient designs critical. Additionally, the growing effects of climate change require the urgency of building homes that can adapt to and mitigate these challenges.

The façade we are designing takes these factors into account and incorporates a combination of advanced thermal insulation, passive solar strategies, and innovative materials to optimize performance. By orienting the façade to the south, we maximize the potential for solar energy gain during Ottawa's colder months. The design allows sunlight to penetrate and warm the interior during the winter. Additional features such as the overhang and low solar heat gain coefficient (SHGC) windows prevent overheating during the summer months. This careful balance ensures year-round comfort and energy savings.

Our approach to sustainable home design prioritizes energy efficiency, environmental responsibility, and occupant comfort. Energy efficiency is achieved through the use of high-R-value insulation, which minimizes heat transfer and significantly reduces heating and cooling demands. Sustainability is at the core of our design, with materials and systems chosen not only for their performance but also for their environmental impact and durability. Comfort is a key outcome of our design. By maintaining stable indoor temperatures occupants can enjoy a more pleasant living environment regardless of the outdoor conditions. This focus on occupant well-being ensures that our design not only meets energy efficiency goals but also enhances comfort quality.

In summary, our south-facing façade design addresses the unique challenges posed by Ottawa's climate while embracing the principles of sustainability and energy-conscious living. By integrating solar energy utilization, advanced thermal insulation, and innovative design strategies, we aim to create homes that are not only resilient to Ottawa's harsh winters but also contribute to a greener, more sustainable future.

Performance Metrics

Our energy saving focused performance metric is to produce a minimum of 500kWh/year of energy from the solar panels on our wall section. Since this is just a small wall section of a building it would be unrealistic to try to produce enough energy from this wall alone to offset all the building energy consumption but we wanted to try to make at least a bit of an impact on the total energy consumption. The average single home in Ontario uses about 1100kWh/year [1], this is why we chose to aim to produce 500kWh/yr of energy from this wall, it will offset about 45% of the total energy usage. If other wall sections of the building have similar energy gains from solar panels the building will surely be on track to becoming net zero.

Our life cycle focused performance metric is to have a minimum wall thermal resistance value of 4RSI. To ensure our building will be able to withstand potentially harsher climates in the future and to save even more on energy costs we wanted to have a higher thermal resistance value than the 3.08RSI required by the National Building Code of Canada [2]. This is why we chose to aim for a minimum value of 4RSI since it is a realistic value we can achieve while still providing plenty of thermal insulation. With this higher insulation our building will be able to save money and energy on heating and cooling loads which overtime can add up to large savings for the building owner. This will also make the house more environmentally friendly since it will require less energy to operate which is crucial to consider nowadays as climate change is a pressing issue and buildings are a large factor in emissions.

Our performance metric that focuses on comfort is to try to reduce the glare experienced by occupants while inside. To achieve this we designed the wall to have 2 separate taller windows as opposed to one large window that covers most of the wall area. Having these 2 separate windows allows for the middle break between them to provide shade to the interior space. This

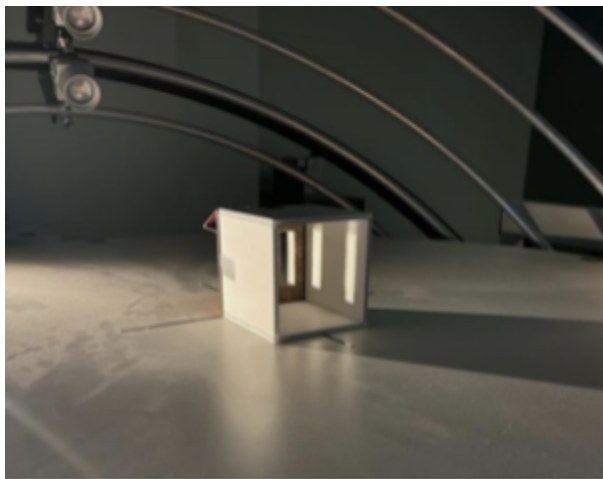


Figure 1: Interior Shading Profile

ensures that at all times it is possible to avoid being in direct sunlight, with one large window it would be hard to avoid glare while inside since there is limited shading. With the two separate windows though we provide a shade that can be used to work in and ensure reduced glare on any screens or surfaces that an occupant needs to use. Looking at Figure 1 below taken during the heliodon test we can see the lighting pattern on the interior wall during the mornings in the winter and the large shaded area provided to prevent glare. Since our model prevented all light from penetrating during the spring and summer from 9am-3pm we don't

have to worry about glare, and outside these hours the shading between the windows would still be there with the same effect as we designed for.

Sustainable Passive Strategies

Ottawa's climate consists of frigid winters and warm summers, therefore any building must be prepared for large temperature variations depending on the season. Winters are particularly cold, meaning it's essential to have energy-efficient insulation to retain as much heat as possible and minimize energy loss. Summers, on the other hand, are the exact opposite, containing boiling heat waves and requiring buildings to manage cooling effectively. As seen in Figure 2 the ground temperatures for Ottawa have been generally increasing as climate change takes its effect. These ground temperatures also fluctuate seasonally, with soil freezing in winter and thawing in spring, which can impact the efficiency of heating and cooling systems. Looking to the future, Ottawa is expected to experience more extreme weather patterns due to climate change, including hotter summers and more intense winters. These shifting climate conditions make energy-efficient home features more critical in lowering heating and cooling costs.

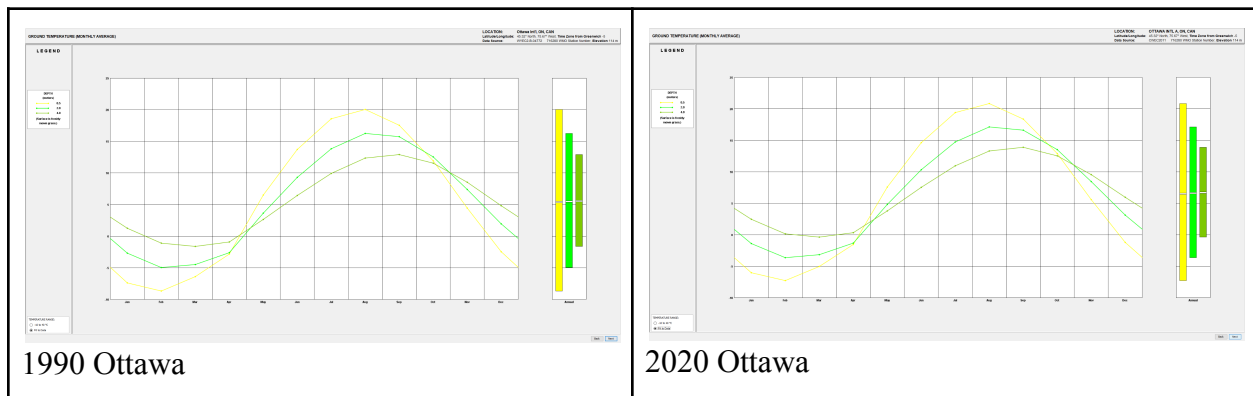


Figure 2: Ground Temperature in Ottawa for 1990 and 2020.

Included in our design are many passive design features that will help in terms of present day requirements while also taking into consideration future climate needs. Starting with the opaque envelope, our wall detail incorporates multiple forms of insulation, ideal for maintaining comfortable indoor temperatures. This combined with our triple-glazed low-E argon vinyl framed windows result in extremely efficient overall thermal performance in both the harsh winters and warm summers. Our overhang was also designed for the varying Ottawa climate, in general letting in the maximum amount of sun during the winter and the minimum during the summer. Meaning the room receives solar heating in the cold winter while staying shaded during the summer months. To complement our thermal performance are PV solar panels for efficient energy production. All these features contribute to our passive system designed for optimal efficiency in the Ottawa climate.

Overall Facade Design

The facade design of our project prioritizes energy efficiency, resilience, comfort and views to



Figure 3: Facade Design

the outdoors. In terms of energy efficiency our facade contains materials that ensure that thermal heat loss is kept to a minimum. Our windows contain two argon layers, this was chosen as argon is denser than air and has a lower thermal conductivity level [11]. The visual design of the facade, containing two narrow windows, ensures that there is minimal heat loss through the glass panes as opposed to one large window. With the combination of these strategies we have prioritized the energy efficiency of our facade.

A second aspect we have prioritized is resilience. With the design of two narrow vertical windows we are able to reduce vulnerabilities due to wind and debris in extreme weather conditions while maintaining structural integrity. Our overhang offers protection from the elements by shielding the wall and windows from heavy rainfall, hail, and direct sunlight, reducing wear and tear. Our design prioritizes durability, safety, and adaptability to environmental challenges.

Our facade design encompasses many aspects that prioritizes comfort. We ensured our design contains operable windows which allows the occupant to customize airflow throughout the space. To ensure visual comfort is met we included two windows, allowing indirect light into the space while still ensuring that the windows are covered during 9am-3pm from the months of May to June. The quality of natural light, with its dynamic changes throughout the day, is gentler on the eyes compared to artificial lighting. Natural light creates a healthier and more sustainable indoor atmosphere.

The dual window design also offers optimal views to the outdoors. While our design contains vertical shading elements we ensured that the side fins are the minimal depth required to shade the windows during peak months. This was calculated to ensure that the fins do not obstruct the view to the outdoors. The side fins were also designed to be operable so that they can lay flat against the facade during winter months, where full solar penetration is desired. Our design contains aspects that prioritize energy efficiency, resilience, comfort and views to the outdoors. This was all achieved by creating a harmonious facade design.

Opaque Envelope

The wall assembly is designed with a total depth of approximately 250 mm, incorporating multiple carefully selected layers, each serving specific functions to ensure optimal performance

in terms of thermal resistance, structural integrity, and weather protection. The vinyl siding, positioned on the exterior, serves as the first line of defense against external weather conditions, including rain, wind, and snow. This layer provides durability, resistance to UV degradation, and low maintenance requirements, making it an excellent choice for Ottawa's harsh climate.



Figure 4: LBNL Therm Results

To prevent moisture from compromising the insulation and structural integrity of the wall, dual vapour barriers are installed between the exterior rigid insulation. These layers act as a critical shield against water vapor diffusion, ensuring condensation does not form within the wall assembly, which could otherwise lead to mold growth or material degradation. The layer of rigid insulation chosen is EPS (Expanded Polystyrene). It is

chosen for its lightweight, high R-value, and excellent moisture resistance, making it a vital component for reducing conductive heat transfer through the wall.

In the wall detail lies the OSB (Oriented Strand Board) layer, which acts as a key structural component of the wall assembly. It provides rigidity and stability to the wall while also offering some degree of thermal resistance, although its primary purpose is to support the other layers. The spray foam insulation, filling the cavities between the wall studs, offers even higher thermal resistance due to its superior insulating properties and ability to expand and seal gaps. Spray foam not only prevents heat transfer but also enhances the airtightness of the wall, minimizing the infiltration of cold air during winter and hot air during summer. This airtight barrier contributes significantly to reducing energy losses and maintaining indoor comfort. On the interior side, 5/8-inch drywall is used to complete the wall assembly. Drywall provides a smooth, fire-resistant, and durable surface for interior finishes while adding a minimal amount of thermal resistance to the overall wall system.

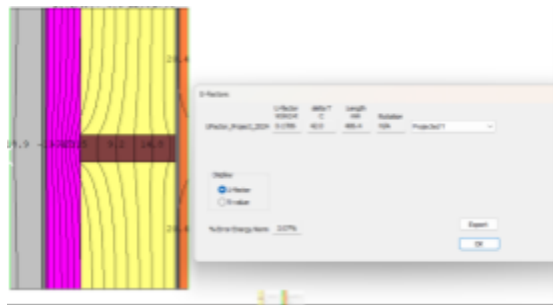


Figure 5: LBNL Therm Results

The overhang design enhances the wall assembly by managing solar heat gain. Constructed with wood supports and finished with vinyl cladding, the overhang blocks the high-angled summer sun to reduce cooling loads while allowing in the low-angled winter sun to maximize passive heating. This south-facing façade was designed for energy efficiency and to suit Ottawa's climate, with a calculated R-value of $5.8631 \text{ m}^2 \cdot \text{K}/\text{W}$. This exceeds

the required ASHRAE standards, with a U-value of $0.14 \text{ W}/\text{m}^2 \cdot \text{K}$.

Fenestration

Triple-glazed low-E argon windows with vinyl framing is the window we choose for Ottawa's

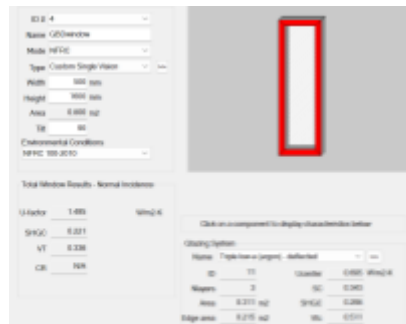


Figure 6 - LBNL Window simulation

varied climate, offering significant energy efficiency, comfort, and durability. With Ottawa experiencing both the extremely cold in winter and sweltering warmth in summer our window must be versatile enough to deal with the differing challenges of each season. The triple glazing provides superior insulation compared to single or double-glazed windows, reducing heat loss in winter and minimizing heat gain during the summer. The simulated u-factor of the window came out as 1.495 W/m²K. A very respectable number that adheres to ASHRAE window requirements and should easily be able to deal with the

aforementioned seasonal challenges.

These well rounded insulation properties help maintain a consistent indoor temperature year-round, ensuring homes stay warm without excessive heating costs in winter, and cool without over-relying on air conditioning in summer. Low-emissivity (low-E) coatings on the glass further enhance energy efficiency by reflecting heat back into the home during cold months and blocking harmful UV rays in the summer [9]. Between the panes of glass is the non-toxic, odorless gas argon, that adds another layer of insulation by slowing the transfer of heat, and improving thermal performance.

This type of window also contains anti-frost and condensation properties, resulting in better

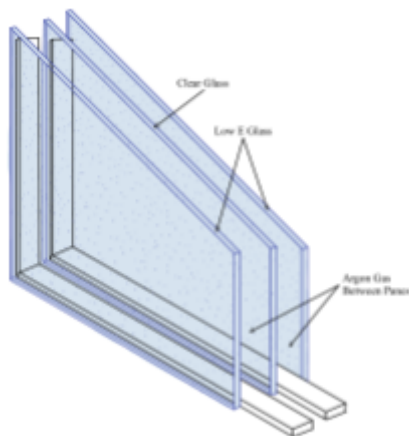


Figure 7 - Low-e triple glazing with argon gas

protection to mold growth and water damage [10]. Vinyl framing complements the glass's energy efficiency by providing excellent thermal resistance. Vinyl is durable and low-maintenance, while being resistant to moisture and rot, which is important in Ottawa's diverse weather conditions. Vinyl frames won't warp, crack, expand or contract with temperature changes, ensuring the window remains sealed and efficient for years. It accomplishes this while improving greatly on thermal insulation compared to an aluminum frame. Vinyl, unlike metal frames, is a poor conductor of heat, which means it maintains more consistent indoor temperatures by reducing thermal bridging.

Overall, triple-glazed low-E argon windows with vinyl framing are ideal for Ottawa's climate, utilising energy efficiency, comfort, and longevity with a minimal environmental impact.

Solar Shading

Our solar shading design offers complete shading between the hours of 9am and 3pm in May, June and July. Several iterations were performed to optimize the solar penetration and shading throughout the year.

The first step taken to optimize the solar shading was to gauge the overhang necessary to shade a 1x1 [m] window from 9am-3pm for May 21st and July 21st. The window would be fully shaded with an overhang depth 0.4 [m]. However, to achieve our daylighting goals and provide a customizable experience for the occupants, a second window was added and the dimensions of each became 0.5x1.6 [m]. Pictured in Figure 8 below is the wall simulation on May 21st, created using the overhang simulator offered by *Sustainable By Design*, where there is solar penetration in one of the windows [5]. An overhang of depth 0.44 [m] was needed to shade the new window height, but our chosen depth of 0.9 [m] was determined after multiplying by a factor of 1.45 to lessen diffuse lighting.



Figure 8 - Overhang shading at 9AM on May 21.

The simulation proved there would have to be multiple shading elements to provide complete shading from May 21-July 21, since it is impossible with the overhang alone regardless of its depth. For that reason, side fins were implemented into the design, providing an element of vertical shading to compensate for the overhang's lack of coverage. Determined through hand calculations using the solar azimuth angle, and confirmed through *Sustainable By Design* and the solar shading simulation feature in *Sketchup*, a side fin depth of 0.18 [m] was found to be the minimum depth to shade the windows. However, the depth of the side fins was increased by a factor of 45% in order to mitigate any diffuse solar energy, resulting in a depth of 0.26 [m]. The calculations to determine the necessary overhang and side fins can be found in Appendix A.



Figure 9 - Shading % with minimum depth (left) & adjusted depth (right)

To optimize the amount of direct sunlight passing through the windows, particularly when temperatures are lower, the side fins are adjustable to offer a greater duration of solar penetration. This allows for an increased amount of solar gains compared to fixed side fins [6].



Figure 10 - Successful Heliodon trials

A physical 1:30 scale of the model was created to measure the solar penetration at various instances using a Heliodon. The windows in the model were expected to be completely shaded from 9am-3pm on May and June 21st, and have near-total solar penetration on December 21st. As shown in Figure 10, the model met all expectations.

Daylighting

The daylighting design was perfected to ensure light exposure and penetration in the winter months while shading in the summer months. While we aimed to reduce the solar heat gain in the summer, we still wanted to allow reflected light to offset the electricity needed throughout the day. This has increased the visual comfort for the occupants, as natural light is much better overall because it releases hormones to make us more alert, productive, and active [7].

The side fins and overhang block the sun when it is the most powerful, resulting in less cooling requirements and energy savings. We calculated the overhang length and width for the summer solstice to find the optimal dimensions to balance out solar heat gains, light penetration, and energy consumption. We wanted to make sure that our design had these three main features resulting in a net reduction in energy consumption.

The side fins are on hinges to allow for them to be positioned in any orientation throughout the year. This has allowed for additional solar penetration in the winter months as the fins will be against the wall, while still maintaining minimal direct penetration in the summer months as they will be adjusted to their optimal position.

Table 1: Electric Lighting Offset

	Winter [lux]	Spring [lux]	Summer [lux]	Fall [lux]
Sunrise	400	400	400	400
During the Day	45,000	90,000	110,000	80,000
Sunset	400	400	400	400

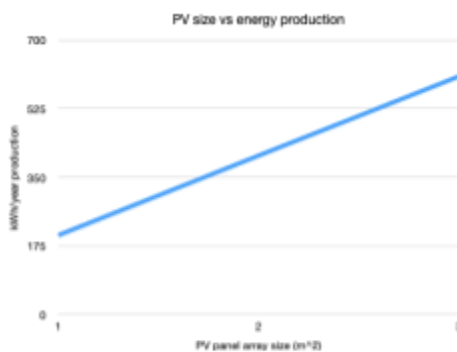
Before / After sun sets		50	50	50	50
Sunset / Sunrise	$DA_{Cont, 500lux}$	$400/500 = 0.8 = 80\%$			
During the Day	$DA_{Cont, 500lux}$	$45,000/500 = 90$	$90,000/500 = 180$	$110,000/500 = 220$	$80,000/500 = 160$
Before / After Sunrise / Sunsets	$DA_{Cont, 500lux}$	$50/500 = 0.1 = 10\%$			

[8]

The above table shows that during the day we will have all of our light needs met. We designed the overhang with the side fins to allow for ample sun penetration during the winter, due to decreased amount of solar lumination. During sunrise, 80% of the required light is supplied naturally through the windows, while during the sunset only 10% of what we need will be supplied. Although, during the summer, most direct natural light is blocked by the side fins and overhang. The indirect sunlight will provide enough lumination throughout the day to meet our 100% offset.

Solar PV

In order to meet our performance metric that states we want to produce a minimum of 500kWh/year of energy from solar we needed to ensure that the solar panels were an adequate size to produce this. The solar panels are to be crystalline silicon panels and placed on the overhang since it is a relatively large surface and will be exposed to sun for most of the year whereas most of the rest of the wall will be shaded at times due to the overhand. We chose crystalline silicon panels since they are highly efficient, well researched and have a long lifespan that could last over 25 years so they will continue to generate energy over a long span of the building's life and reduce maintenance cost [4] .



Graph 11: PV array size and energy production

Putting the solar panels on the overhang does limit us to a finite amount of space to work with so we must ensure it is enough space. The maximum space we have to work with on the overhang is about 3.24m^2 so to ensure this is enough space we used the online software PVwatts to calculate the minimum space required to produce this

500kWh/year we want. First when using PVwatts we have to enter the location we are in which is Ottawa, Ontario, then the azimuth which since we are facing south is 180° and the tilt which is based on our overhang is 33.69° . Using the provided assumptions for the PV panels on PVwatts we can provide the size of the system which was calculated from the array size (See Appendix A). We first started by finding a relationship between the array size and the energy output so we can then figure out what the minimum array size to provide 500 kWh/year would be. We started by using the array sizes of 1 m^2 , 2 m^2 and 3 m^2 , the results can be seen in Graph 11 which shows a linear relationship between size and energy production. Using this linear relationship we used linear interpolation to find at what array size would produce 500 kWh (See Appendix A). We found that to produce our desired 500 kWh/year the solar array would need to have an area of at least 2.47 m^2 , since this is less than the area of the overhang we are able to meet our requirement. Since our overhang is larger than the minimum area required we can design the solar array to cover the entire overhang to be able to produce even more than our designed 500 kWh to try to make our building even closer to net zero.

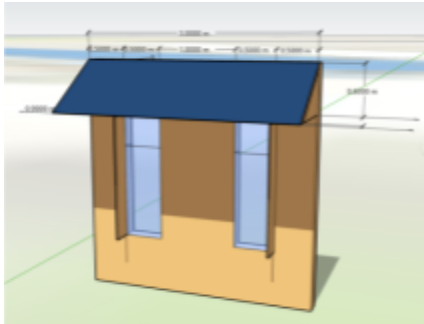


Figure 12: Planned Solar Arrangement

Covering the entire overhang with solar panels which would be an area of 3.24 m^2 we can generate a maximum of 632 kWh/year which exceeds our performance metric and can provide even more clean energy to the building, see Figure 12 below for proposed solar panel placement. (See Appendix A for more calculations and results).

Design Brief

The design for our south facing facade went through 3 iterations. The first design our team created featured a large central window and a 39 degree overhang spanning all 3 meters of the facade. This initial design was modeled and we ran a number of simulations to see how this design performed. It was evident that we needed to fix our design as the overhang was not



Figure 13: Iteration 2 Design

efficient in completely blocking solar penetration during the times of 9am to 3pm for the months of May to July. We also found that with one large window we had a high amount of heat transfer due to the large surface area of the window. We quickly scrapped this design once we realized it would not perform adequately for the helidon testing.

The second iteration of our design incorporated two narrow windows and an overhang spanning the width of the facade set at an angle of 45 degrees. We wanted to challenge ourselves with the

design and decided to have two windows instead of one. This decision is also based on a number of factors we decided would be more ideal for our design. Two operable smaller windows provide greater control over airflow patterns into the space. This window design also has a smaller surface area than our first iteration which means less heat transfer through the glass. This can help maintain indoor temperatures, reducing heating or cooling demands. Two small windows strategically placed can also provide better distribution of natural light, reducing the need for artificial lighting without causing overheating from direct sunlight. When we modeled our second iteration and ran it through simulations to see if the overhang shaded the windows during the times of 9am to 3pm for the months of May to July, we found our design to be unsuccessful.



Figure 14: Final Design

For our third iteration of the design we kept the overhang width and depth as well as the dual windows. To ensure our design passed the required shading times we added vertical shading elements. As shown in the adjacent figure with the addition of the side fins we achieved zero solar penetration from 9am- 3pm for the months of May to July. To calculate the minimum depth of the side fins we used solar angles in accordance to the time of day required and found the minimum depth to be 0.26M. We wanted to ensure that we did not sacrifice the view of the outside and made our side fins to be adjustable so that they can lay flat during colder months to allow additional solar penetration while still ensuring that the warmer

months are fully shaded. Our final design ensures total penetration for December and zero penetration for May to July.

2D and 3D Renderings

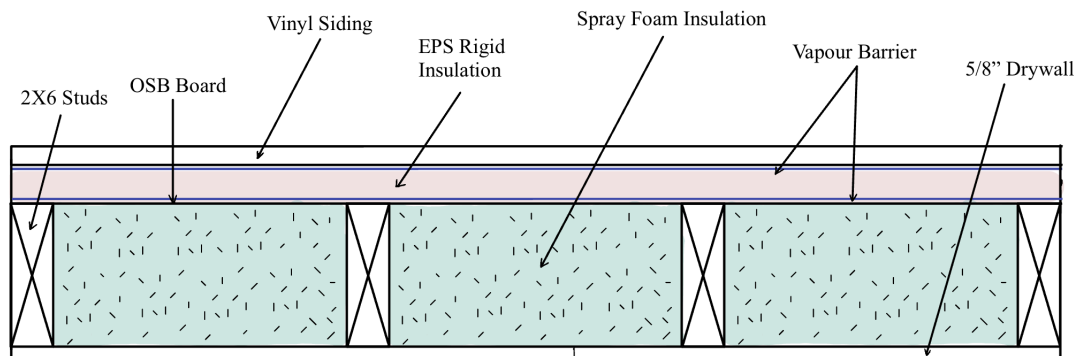


Figure 15: 2D Wall Section Showcasing Materials Chosen

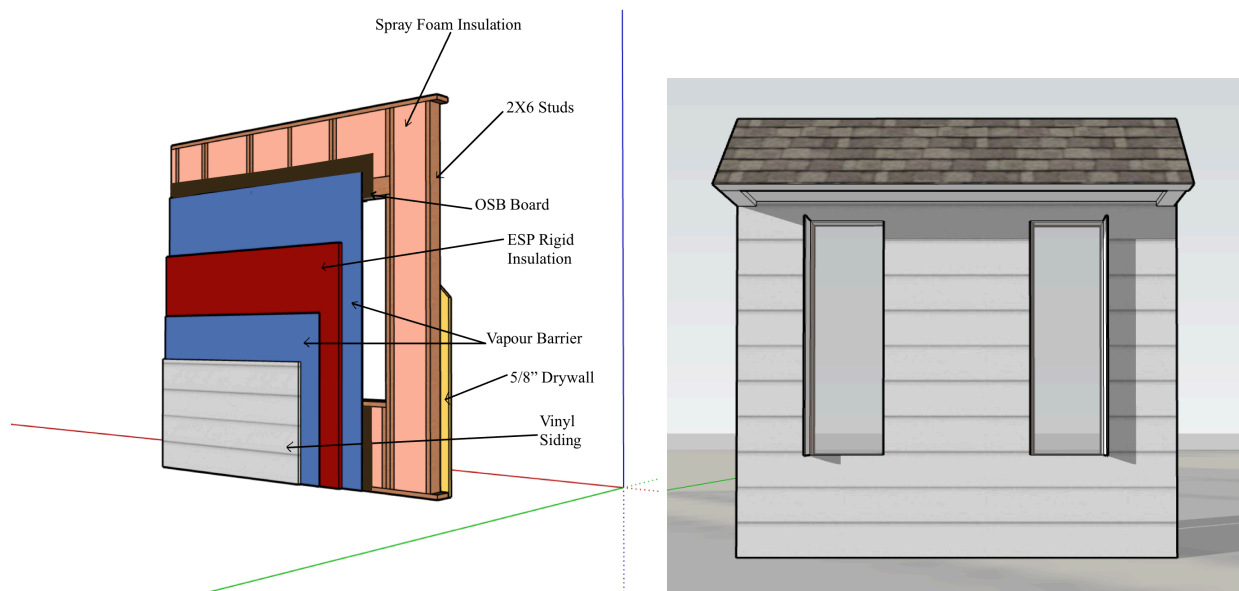


Figure 16: 3D Wall Section

Conclusion

Our façade design for this project integrates advanced thermal insulation, passive solar strategies, and innovative materials to enhance performance. The facade was designed to have complete solar penetration between the hours of 9am to 1pm for the month of December, and zero solar penetration between the hours of 9am to 1pm for the months of May and July. This system allows for solar heat gain in the cold winter months and cool shading during the summer, improving thermal performance throughout the year.

As the facade is oriented to the south, it maximizes solar energy gain during Ottawa's cold winters. Solar energy is produced through the usage of PV panels mounted atop the overhang that can produce 500kWh/year of energy. Our sustainable approach prioritizes energy efficiency, environmental responsibility, and occupant comfort. Our chosen high-R-value insulation exceeds ASHRAE guidelines while reducing heating and cooling demands. The chosen Triple-glazed low-E argon windows with vinyl framing also assist in the thermal performance of the wall.

In summary, this south-facing façade design addresses Ottawa's climate challenges while promoting sustainability. By combining solar energy use, thermal efficiency, and innovative design we have created a resilient and energy efficient building.

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Appendix

Appendix A: Calculations and Results

PV panel size

$$\text{Size (kW)} = \text{Array Area} \times 1 \text{ kW/m}^2 \times \text{Module Efficiency}$$

$$1 \text{ m}^2 \times 1 \text{ kW/m}^2 \times 0.16\% = 0.16 \text{ kW}$$

$$2 \text{ m}^2 \times 1 \text{ kW/m}^2 \times 0.16\% = 0.32 \text{ kW}$$

$$3 \text{ m}^2 \times 1 \text{ kW/m}^2 \times 0.16\% = 0.48 \text{ kW}$$

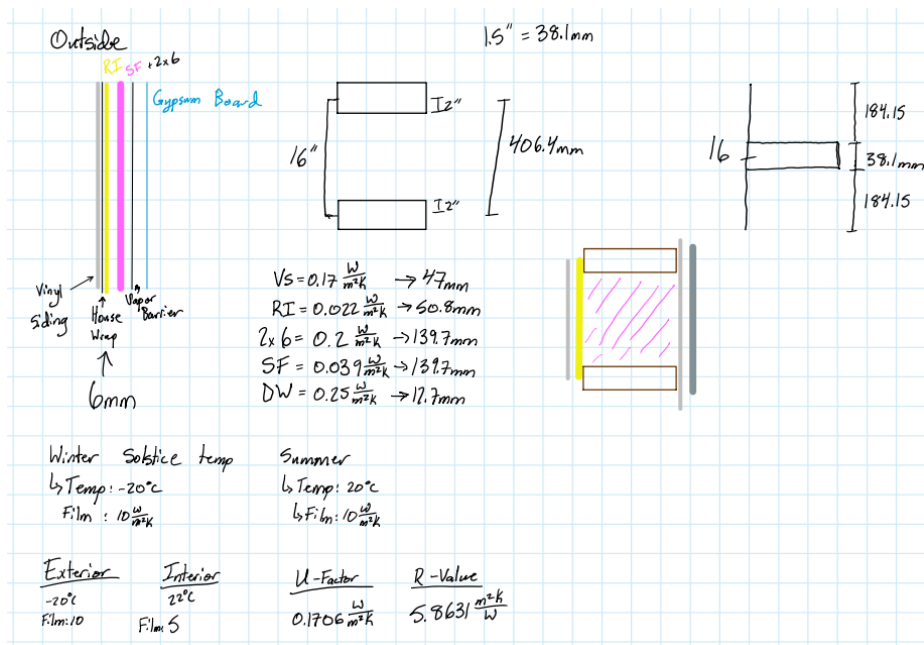
PV production linear interpolation

$$y = y_1 + \frac{(x - x_1)(y_2 - y_1)}{x_2 - x_1}$$

interpolate between
 $x = 2 \text{ m}^2, y = 405$ and
 $x = 3 \text{ m}^2, y = 607$

$$y = 2 + \frac{(500 - 405)(3 - 2)}{607 - 405}$$

$$y = 2.47 \text{ m}^2$$



$$Q_{\text{sol}} = \text{Area} \cdot [SHGC \cdot I_{\text{solar}} + U (T_{\text{out}} - T_{\text{in}})]$$

★ Assumptions for Average sunny day in Ottawa winter

$$= 0.8 \times [0.221 \times \underline{300} + 1.495 (\underline{-18 - 20})]$$

$$= 7.542 \text{ W}$$

[illegible]

PVWatts Calculator

2024-12-02 10:02 AM PST

RESULTS

405

Cashier Photovoltaic system performance estimates generated by PVWatts[®] include solar resource, meteorology, and equipment, and use other system characteristics (such as panel type, inverter efficiency, etc.) to estimate system efficiency. These performance metrics, both field and array, represent average, year-to-year performance for the system based on the weather data and equipment characteristics used in the model. These values are provided for the user to compare with other system performance metrics and to provide a basis for system design and optimization.

The weather data used is based on 30 years of actual weather data at the given location and is available in graphic or tabular form. The weather data is provided in the weather data file, which is available in the weather data file, which is available in the weather data file.

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Month	Solar Radiation (kWh/m ² /day)	AC Energy (kWh/day)
January	2.83	25
February	4.05	31
March	5.55	48
April	5.29	40
May	5.47	41
June	5.37	38
July	5.80	42
August	5.65	41
September	4.65	33
October	3.48	27
November	2.86	23
December	2.40	20
Annual	4.45	406

Location and Station Identification

Requested Location	ottawa
Weather Data Source	Lat/Lon: 45.41°, -75.7° 0.0 mi
Latitude	45.41° N
Longitude	75.7° W

PV System Specifications

DC System Size	0.32 kW
Module Type	Standard
Array Type	Fixed (open rack)
System Losses	14.08%
Array Tilt	33.69°
Array Azimuth	180°
DC to AC Size Ratio	1.2
Inverter Efficiency	96%
Ground Coverage Ratio	0.4
Albedo	From weather file
Bifacial	No (0)

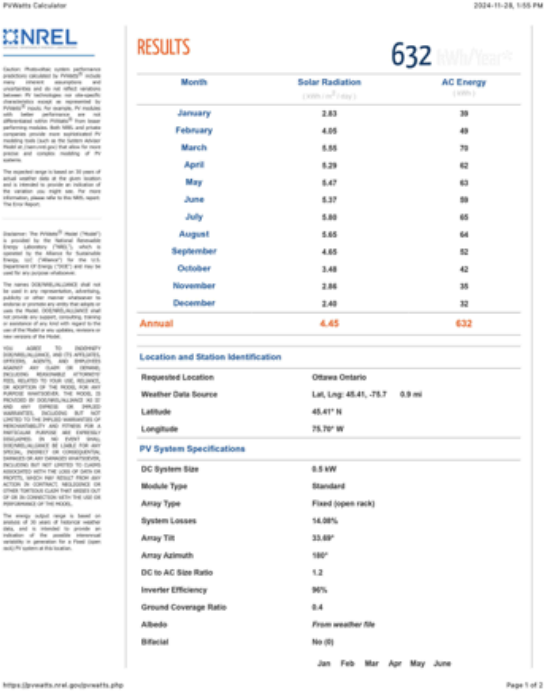
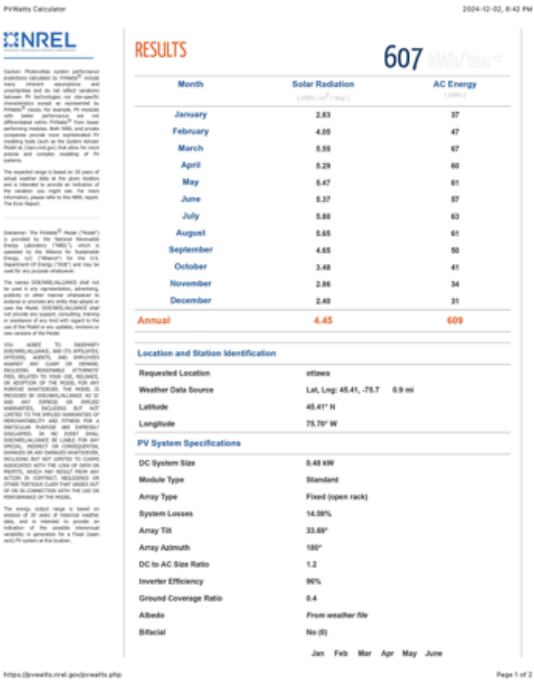
Jan Feb Mar Apr May June

https://nrel.gov/pvwatts/

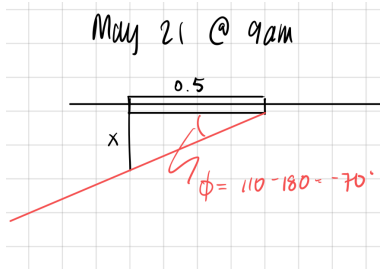
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Profitability Calculator	2024-12-02, 8:41 PM						
Monthly Insurance Loss	July	0%	0%	0%	0%	0%	0%
	Aug	0%	0%	0%	0%	0%	0%
	Sept	0%	0%	0%	0%	0%	0%
Performance Metrics							
DC Capacity Factor	14.4%						

PV Watts Calculator		2024-12-02, 8:43 PM
Monthly Irradiance Loss	0% July 0% Aug 0% Sept 0% Oct 0% Nov 0% Dec	0%
Performance Metrics		
DC Capacity Factor	14.4%	



Solar Shading Calculations:



OVERHANG (profile angle $\Omega=\alpha$)		Formulae
Profile angle	1.221730476	$=\text{Atan}(\tan(L8)/\cos(L12))$
Depth required (m)	0.6187493983	$=(1.6+0.1)/\tan(B35)$
Increase by 45% (mitigate diffuse)	0.8971866275	$=B36*1.45$

SIDE FINS		Formulae
Critical angle (may -> 70 deg)	0.3490658504	$=\pi/2-70*\pi/180$
Depth required (m)	0.1819851171	$=0.5*\tan(B26)$
Increase by 45% (mitigate diffuse)	0.2638784198	$=B27*1.45$